Global Bike Inc.

Data Analyst Onboarding & Setup Guide

Data Science Team

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1 Welcome to Global Bike Inc.

Welcome to the Data Science team at Global Bike Inc. (GBI)! We are thrilled to have you on board.

As a Business Analyst, your role is crucial in transforming our raw enterprise data into actionable strategic insights. In your first project, you will dive into our SAP ERP data to predict Customer Lifetime Value (CLV) and segment our customer base.

This document serves as your onboarding guide. It will walk you through the technical setup required to access our cloud environments and the data you need to start your analysis. Don't worry if you haven't worked with these tools before—this guide is designed to get you up and running step-by-step.

Let's get started!

2 Step 1: Download your Workspace

All the files you need for this case study (the interactive Jupyter Notebook, the theoretical reference material, and the datasets) are hosted on GitHub.

- Navigate to the GitHub repository: `https://github.com/josephkauermann/Business-Analytics-Teaching-Material.git`

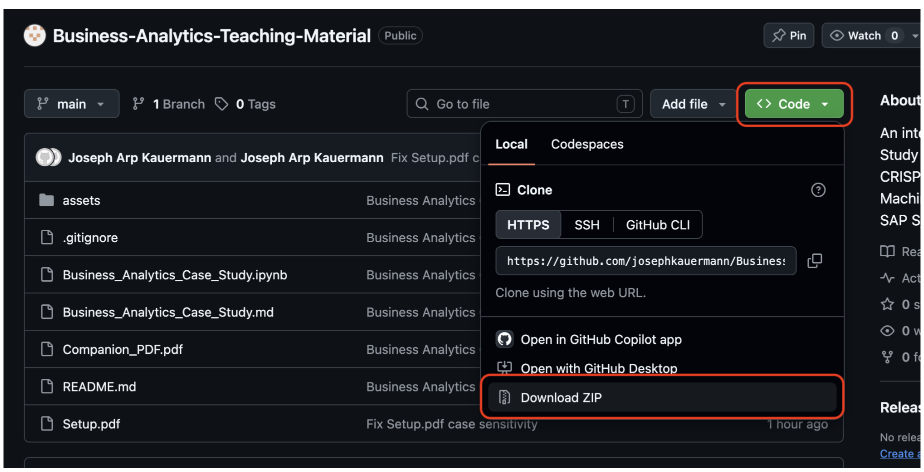

Figure 1: Download the ZIP file by pressing on < > **Code**. Then click on **Download ZIP**

.

- **Important:** The repository contains a *Companion PDF*. Keep this file open locally on your computer, as the Jupyter Notebook will frequently reference it for theoretical background.

3 Step 2: Watsonx.ai Environment Setup

To ensure everyone has the same computational resources and to avoid local installation issues, we use IBM watsonx.ai as our cloud development environment.

- Go to `https://eu-de.dataplatform.cloud.ibm.com/registration/stepone?context=wx&preselect_region=true` and create a free account (or log in if you already have one).
- Once logged in, navigate to the watsonx.ai studio and create a **New Project**.

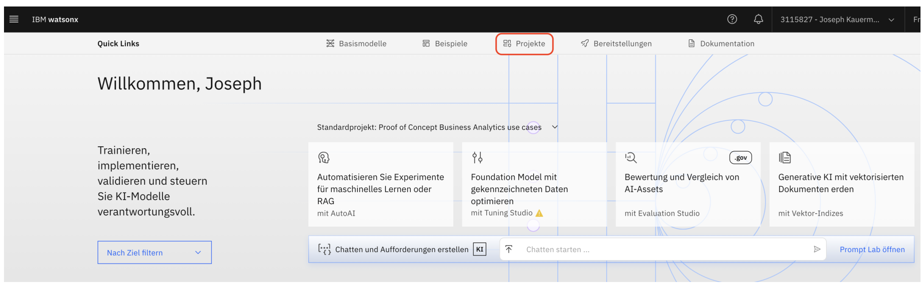

Figure 2: Find the Project tab on the home view. You can also always find your project through the Navigation Menu on the very top left corner (click it to expand, and you will see **Projects**)

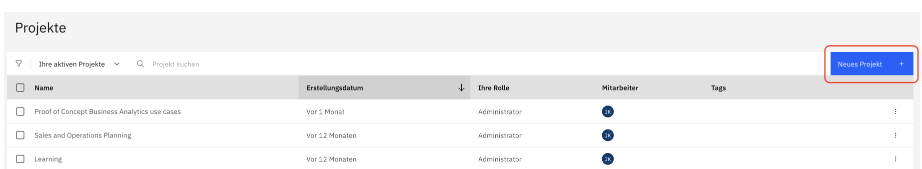

Figure 3: Create **New** Project and choose a name. Then press on **Create**

- **CRITICAL STEP:** You must link a Machine Learning Service to your project. Go to the *Manage* tab → *Services & Integrations* → *Associate Service*. Select the "Watson machine Learning" tile. Select the free Lite plan and create the service. If you skip this step, the AutoAI training later in the case study will fail with a 500 Error ("Service instance not found").

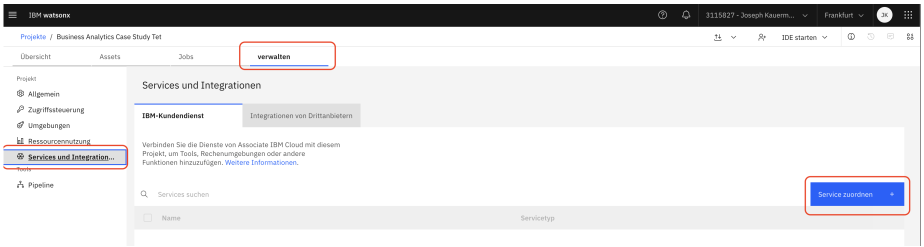

Figure 4: Click on **verwalten**, then on **Services and Integrationen** and then on **Sercice zuordnen**

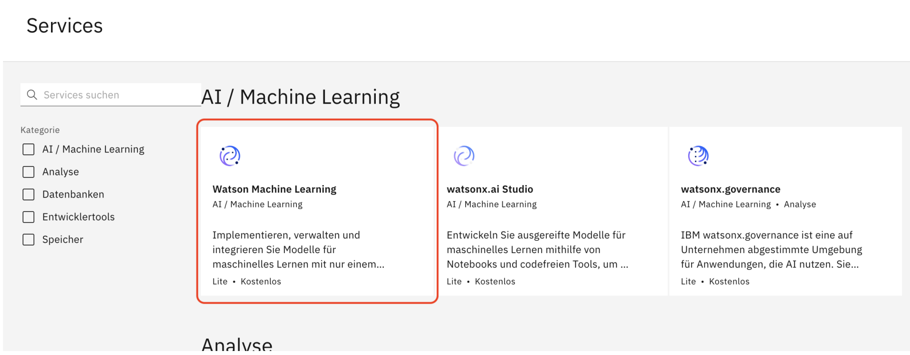

Figure 5: Choose the Service **Watson Machine Learning**

- Note: When creating a new project, watsonx.ai will require you to associate an IBM Cloud Object Storage instance. If you do not have one, simply follow the on-screen prompt to create a free 'Lite' plan instance of Cloud Object Storage, associate it with your project, and click **Create**.

4 Step 3: AutoAI Credentials

Our Jupyter Notebook will communicate with the AutoAI service via an API. You need to generate your personal credentials.

- **API Key:** Navigate to your IBM Cloud profile settings and generate a new API Key. Save this key somewhere safe.

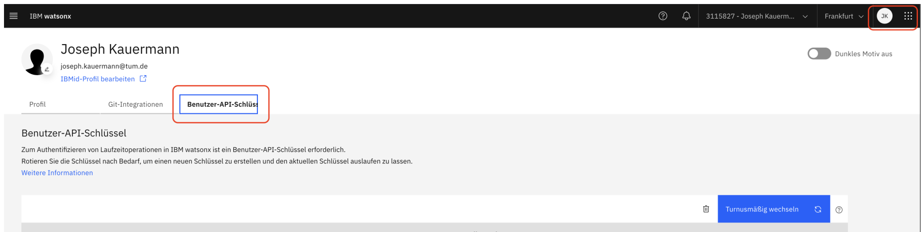

Figure 6: Navigate to your profile settings to create an API key. Write it down for later.

- **Project ID:** Go back to your watsonx project. Under the *Manage* tab → *General*, you will find your Project ID.

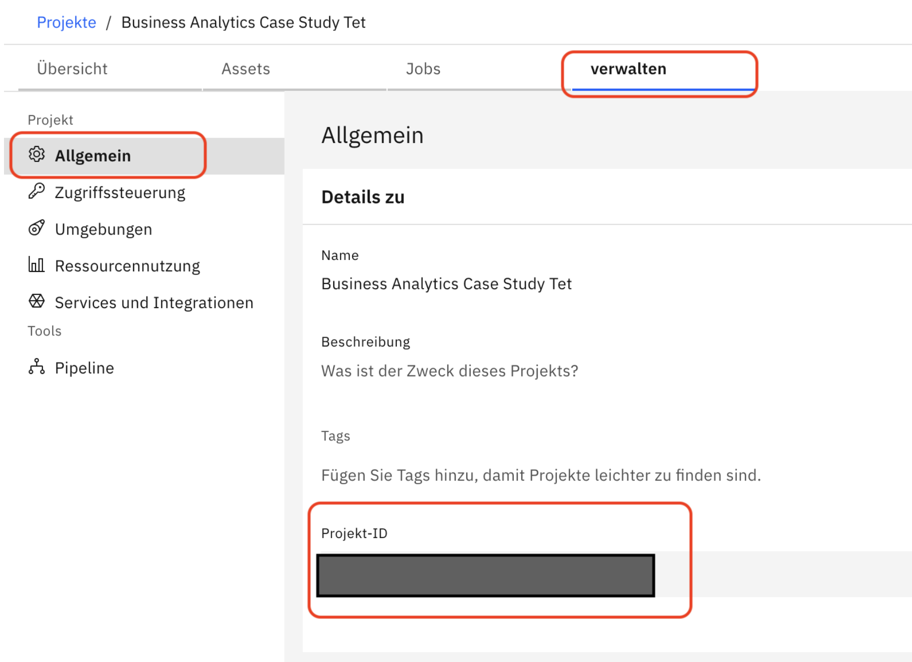

Figure 7: Find your Project ID and write it down somewhere. You will need it later to complete the exercises in the Notebook

- **Region:** When configuring the API in the notebook, ensure you select the correct region (e.g. Frankfurt) where your watsonx instance is hosted. If the region is incorrect, your credentials will be sent to the wrong URL and the connection will fail.

5 Step 4: Launching the Notebook

Now that your environment is ready, it's time to launch the workspace.

- In your watsonx project, go to the *Assets* tab.

Figure 8: Click on Assets and then import Datasets through **Assets importieren** and after that import the Jupyter Notebook through **Neues Asset**

- Upload the Jupyter Notebook `Business_Analytics_Case_Study.ipynb` from your extracted GitHub folder.

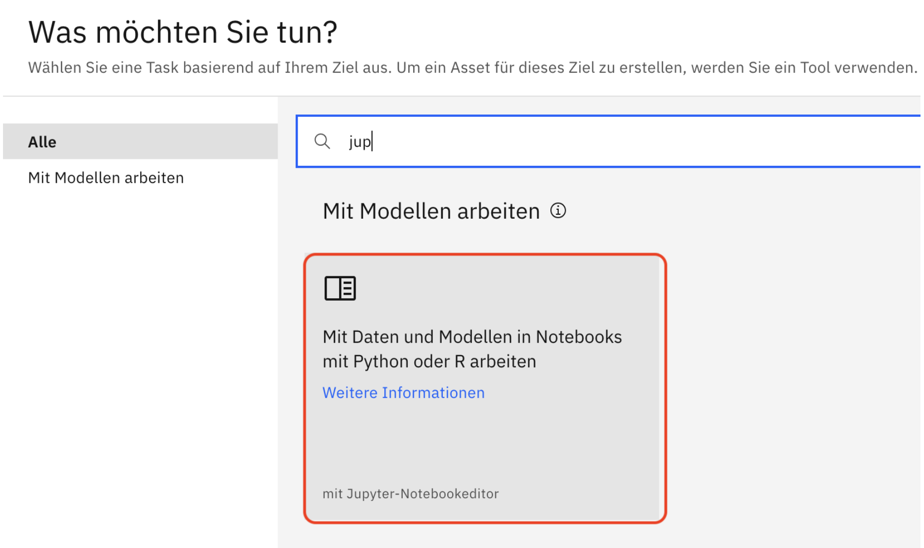

Figure 9: Use the search funcion and select the notebook option. Then choose the Jupyter notebook

- Upload the entire content of the `assets/` folder (all CSV files and the PNG image) into the data assets section.

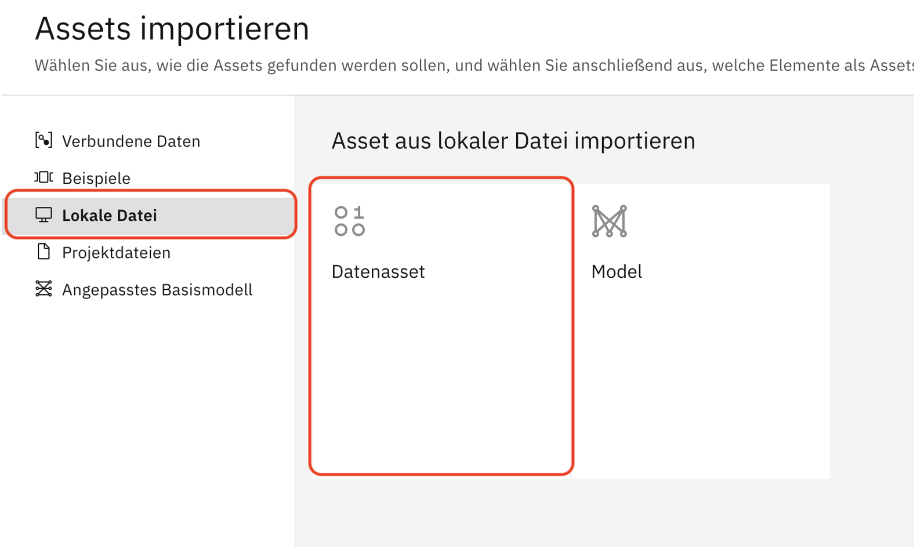

Figure 10: Click on **Lokale Datei**. Then select **Datenasset**. Find the .csv files from the github upload them

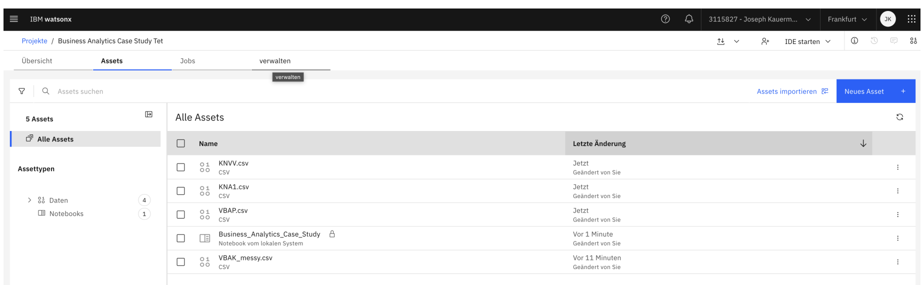

Figure 11: Afterwards, your project should look like this. To open the notebook simply click it.

- Open the notebook. On the top right you might have to press on the **Lock** and **Pen** symbols to unlock the project and to enable editing mode. You are now ready to start coding!

6 Step 5: AI Tutor

Paste this prompt into your preferred AI chatbot (like ChatGPT or Claude) to initialize its role as your personalized AI tutor.

Note: For the best experience, also upload the Case Study Markdown file

(Business_Analytics_Case_Study.md) into the chat so the AI has full context of your exercises!

Copy & Paste the following text:

> "Your role for this chat session is to act as my personalized AI tutor. I am a university student working through a Business Analytics Case Study in a Jupyter Notebook. You are expected to strictly follow this set of constraints:
>
> 1. INITIALIZATION: Begin our conversation by asking me exactly two short questions to estimate my current level of understanding: - 'How comfortable are you with Python and the Pandas library (Beginner, Intermediate, Advanced)?' - 'What is your prior knowledge of Machine Learning concepts?' Wait for my answer before proceeding. Adapt your future explanations based on my skill level.
>
> 2. TUTORING APPROACH (SOCRATIC METHOD): Whenever I have a question or ask for help with a code cell, DO NOT just give me the finished code or the final answer. Your goal is to maximize my learning effect. - Ask guiding questions that nudge me in the right direction. - Keep explanations high-level and business-focused. Explain WHAT a function does and WHY we need it for our specific goal, but do not dive into deep technical details or computer science theory unless I explicitly ask. - Encourage me to figure out the missing pieces of the code myself (filling in the blanks).
>
> 3. FRUSTRATION PREVENTION: While you should not give away the solution immediately, avoid endless loops of guessing. If you notice that I am stuck, frustrated, or my nudged attempts are not leading anywhere, you are allowed to provide the solution. If you provide the solution, always explain line-by-line why it works.
>
> 4. CONTEXT: I have attached the Jupyter Notebook we are working on. When I ask for help, I will refer to the specific exercise or code cell number. Use the attached file to understand the context of my problem."

7 Step 6: SAP RPT-1 Setup

In the later stages of this project, you will interact with SAP's Tabular Foundation Model (RPT-1). **Note:** You do not need to complete this setup right now. You can start working on the Jupyter Notebook and return to this step later when the notebook explicitly instructs you to do so.

- Navigate to the RPT-1 interface: `https://rpt.cloud.sap/dashboard`
- Log in using either your Google account or your SAP User ID.

- You will upload the pre-processed dataset (which our Jupyter Notebook will generate automatically for you) to explore its capabilities.

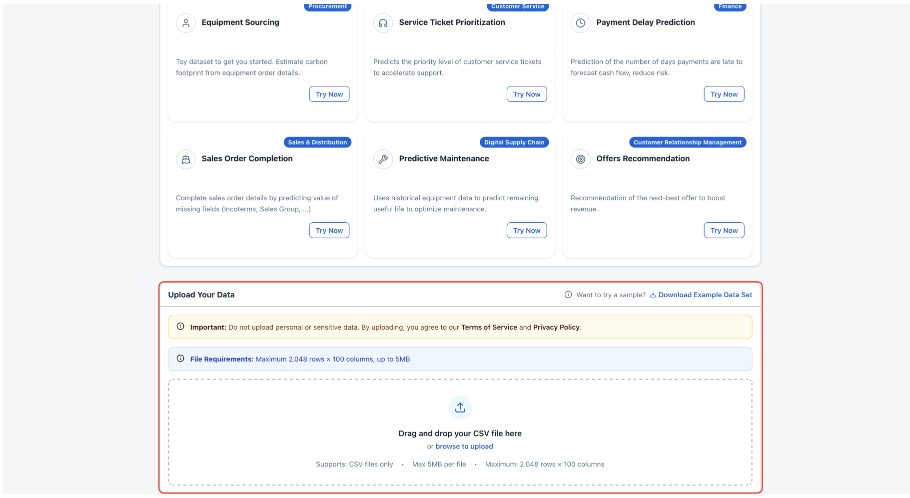

Figure 12: To upload your data you have to scroll down until you see the **Upload Your Data** section